One important difference is that with k-means we had an underlying distribution and choosing the points such that they were uniformly distributed with respect to this distribution was harder and needed particular caution.

For graph embedding, we can draw our points from a finite portion of the plane, typically a rectangle[7], and those points will determine the number of edge intersections.

Let's define the problem more formally now:

Given a simple graph $G=(V,E)$ with $n=|V|$ vertices and $m=|E|$ edges, we need to draw n random points from a finite rectangle whose corners[8] are $(0,0)$ and (W,H), so that:

1. Each point (x,y) is such that $0 \leq x < W$, $0 \leq y < H$;
2. Edges will be drawn as straight-line segments;
3. Vertices will be drawn as points (or circles) centered at those n points;
4. No two vertices can be assigned to the same center, so given two points $(x1, y1)$ and $(x2,y2)$ either $x1 \neq x2$ or $y1 \neq y2$;
5. No vertex can lie on the path of an edge;
6. We assume G has no loops[9]: if it had, we could always draw loops without intersecting any other edge.